GEHU/02E/9.1.3

Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act 1956

End Term (Even) Semester Regular/Back Examination May 2026

Roll no.............

Name of the Program and semester: B. Tech - CSE VI
Name of the Course: Computer Networks - II
Course Code: TCS614

Time: 3 hours | Maximum Marks: 100

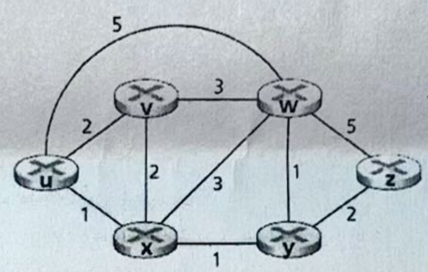
Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2×10=20 Marks)

a) Classify routing algorithms. Describe the advantages and limitations of each class of routing algorithms. [CO1]

b) Apply Link-State routing algorithm on the following graph and find the least cost path and forwarding table for node 'v'. [CO1]



c) Describe the inter-AS routing in BGP and explain how each router's routing table entries are updated. Give an example. [CO1]

Q2. (2×10=20 Marks)

a) Find the code word for $G(x) = x^3 + x^2 + 1$ and the information bits 11001110011111 using cyclic redundancy check. [CO2]

b) Draw the frame format used in IEEE 802.3 Ethernet and explain the functions of each field present in the format. [CO2]

c) Explain Virtual LAN and describe various uses of VLAN in practice. Explain the process of VLAN formation using a single Switch. [CO3]

Q3. (2×10=20 Marks)

a) Suppose an analog audio signal is sampled 16,000 times per second, and each sample is quantized into one of 1024 levels. What would be the resulting bit rate of the PCM digital audio signal in MBps? [CO4]

b) Name and explain limitations of best effort service in streaming live audio and video? How jitter at the receiver for audio could be removed? [CO4]

c) There are two types of redundancy in video. Describe them, and discuss how they can be exploited for efficient compression. [CO4]

End Term (Even) Semester Regular/Back Examination May 2026

Q4. (2×10=20 Marks)

a) Draw the architecture of Software Defined Networking and explain the functions of control plane. [CO5]

b) Describe the working of OpenFlow protocol in SDN by help of important messages flowing from SDN controller to the controlled switch and vice-versa. [CO5]

c) Differentiate between pre-SDN and SDN network environment. Also explain the limitations of SDN. [CO5]

Q5. (2×10=20 Marks)

a) How does a parent process pass a connection to a child, and what are the specific responsibilities of each process regarding the close() function to ensure resources are not leaked? [CO6]

b) In an abstraction using TCP and UDP protocols, describe the use of socket addresses in establishing a client-server connection. [CO6]

c) In client-server architecture, explain the application of bind, listen, accept, fork and exec functions in socket programming.

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